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GNIOT Institute of Management Studies – PGDM (Batch 2021-23)

(TRIMESTER – III) END TERM EXAMINATION, APRIL - 2022

Marketing Analysis for Business Decisions

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 100]

INSTRUCTIONS:

- (i) **There are THREE sections in this question paper.**
- (ii) **Attempt questions from each section as per the directions.**
- (iii) **Make suitable assumptions wherever necessary.**

SECTION-A

(Short Answer Type Questions, do not exceed 100 words)

Q.1. Attempt all parts of the following:

(10×3=30)

- (a) Differentiate between the regression and classification model.
- (b) What is multi Collinearity and why it should not be in your data?
- (c) Differentiate between correlation and covariance.
- (d) What is churn rate?
- (e) Differentiate between time series and cross section data.
- (f) Define collaborative filtering in recommender system and what is its main principle?
- (g) What do you mean by correlation? Mention 4 uses of it...
- (h) Differentiate between Trend and Seasonality.
- (i) What is the 3rd step in crisp dm . kindly define them..
- (j) Suppose in a dataset there are 200 rows then how many cluster can we create for this?

SECTION-B

(Long Answer Type Questions)

Attempt all questions of the following:

(4×10=40)

Q.2. (a) Discuss the importance of business analytics

OR

(b) Explain all kinds of model based approaches to forecast the time series data.

Q.3. (a) “ Improved segmentation can lead to significantly improved marketing effectiveness” Justify.

OR

(b) Explain the apriori algo in association rule with suitable example

Q.4. (a) Explain the first and second business moment decision.

OR

(b) What are the different steps that you need to perform while doing clustering

Q.5. (a) Write down the steps that you follow while building your supervised machine learning model

OR

(b) What are the key factors that predict customer churn?

SECTION-C

Q.6. Attempt all parts of the following:

(2×15=30)

6(a) Below are the scores obtained by a student in tests 34,36,36,38,38,39,39,40,40,41,41,41,41,42,42,45,49,56

- i) Find mean, median, variance, standard deviation.
- ii) What can we say about the student marks?
- iii) What is the nature of skewness when mean, median of data are equal?
- iv) What is the nature of skewness when mean > median?
- v) What is the nature of skewness when median > mean?
- vi) What does positive kurtosis value indicates for a data?
- vii) What does negative kurtosis value indicates for a data?

6(b) Answer the following three questions based on the histogram below.

- (i) Where would the mode of this dataset lie?
- (ii) Comment on the skewness of the dataset.
- (iii) Suppose that the above histogram and the box plot in question 2 are plotted for the same dataset. Explain how these graphs complement each other in providing information about any dataset.

